Prodigy Anesthesia Quick Review

Hypoaldosteronism/Hyperaldosteronism

What is the primary function of aldosterone?	Aldosterone promotes the reabsorption of sodium and the secretion of potassium in the renal tubule. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 925.*
How does an excess of aldosterone affect serum sodium and potassium levels?	An increase in the level of aldosterone can decrease the level of sodium excreted in the urine significantly. It will also result in the dramatic loss of potassium into the urine. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 925.*
How does a lack of aldosterone affect serum sodium and potassium levels?	A lack of aldosterone can result in the loss of as much as 10-20% of the total body's sodium content per day. Simultaneously, potassium will be conserved. The total lack of aldosterone secretion will result in death in as little as three days. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 924-925.*
What ventilatory concerns should you have when anesthetizing a patient with hyperaldosteronism?	Hyperventilation could worsen the hypokalemia associated with hyperaldosteronism. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 397.*
What is Conn's syndrome?	Conn's syndrome is another name for primary hyperaldosteronism. Primary aldosteronism is characterized by an excess of aldosterone due to a functional tumor. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 397.*
What conditions are associated with primary hyperaldosteronism?	Hyperaldosteronism may occur with pheochromocytoma, hyperparathyroidism, and acromegaly. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 397.*
What is secondary hyperaldosteronism?	Secondary hyperaldosteronism occurs when aldosterone levels are elevated as a result of increased renin levels. Renovascular hypertension is a common cause of secondary hyperaldosteronism. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 397.*
What are the signs and symptoms of hyperaldosteronism?	The symptoms are often nonspecific. Systemic hypertension from sodium retention may cause headaches. Hypokalemia may produce polyuria, skeletal muscle weakness, nocturia, and muscle cramps. It eventually results in a hypokalemic metabolic alkalosis. Hypomagnesemia and abnormal glucose tolerance may also be present. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 397.*

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What signs are suggestive of hyperaldosteronism?	Hypokalemia in the presence of hypertension is suggestive of hyperaldosteronism (if potassium wasting diuretics are not being administered). *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 397.*
How can a diagnosis of hyperaldosteronism be ruled out?	An aldosterone level of 9.5 ng/dL after a saline infusion is used to eliminate hyperaldosteronism as the cause of hypokalemia and hypertension. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 397.*
How can hyperaldosteronism affect the use of nondepolarizing muscle relaxants?	The hypokalemia associated with hyperaldosteronism can result in skeletal muscle weakness and potentiation of nondepolarizing muscle relaxants. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 397.*
What is the most common cause of primary hyperaldosteronism?	The most common cause of primary hyperaldosteronism is unilateral adenoma in the adrenal gland. About 25% of these patients, though, may exhibit bilateral adenoma. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 422.*
What are the cardiac considerations for a patient with hyperaldosteronism?	Patients with hyperaldosteronism have a high incidence of ischemic heart disease. A careful preoperative evaluation and close monitoring during anesthesia should be performed. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 422.*
How might the presence of hyperaldosteronism affect your choice of volatile anesthetic?	Hyperaldosteronism can result in hypokalemic nephropathy and polyuria. In this scenario, the potentially negative effects of sevoflurane on the kidneys might preclude its use. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 397.*
Should patients undergoing surgery for excision of an aldosteronoma receive supplementary exogenous corticosteroids?	A patient undergoing surgery for an isolated adenoma of the adrenal gland that is secreting aldosterone probably will not need exogenous corticosteroids. A patient presenting for excision of bilateral aldosteronomas, however, will likely need supplementation with corticosteroids. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 397.*
How is hyperaldosteronism treated?	Hyperaldosteronism is treated with potassium replacement and an aldosterone antagonist such as spironolactone. Bilateral adrenalectomy may be required if the cause is multiple, bilateral, aldosteronomas. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 397.*

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How is hypoaldosteronism treated?	Patients are administered fludrocortisone and placed on a high sodium diet. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 398.*
What lab findings are associated with hypoaldosteronism?	Hyperkalemia that occurs in a patient with normal renal function suggests hypoaldosteronism. Hyponatremia is also present in these patients. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 397-398.*
What are the cardiac symptoms of hypoaldosteronism?	Hyperkalemia due to hypoaldosteronism will often produce heart block. Hyponatremia may produce orthostatic hypotension. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 397-398.*
What are the possible causes of hypoaldosteronism?	Hypoaldosteronism can occur from adrenalectomy, prolonged heparin administration, diabetes, and renal failure or can occur as a congenital condition. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 423.*
How are plasma renin levels affected by hypoaldosteronism?	Hypoaldosteronism is associated with decreased plasma renin levels. Renin levels also fail to rise in response to sodium restriction and diuretic administration. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 423.*
How is acid-base status affected by hypoaldosteronism?	Hypoaldosteronism produces hyperkalemic metabolic acidosis. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 423.*

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